

Software Flow

Smart Cold-Chain System — Farm-to-Market Closed Cold-Chain | Firmware, Backend & Dashboard Architecture

1. Overview

The software stack has four layers: on-crate firmware (closed-loop control), a communication layer (crate/warehouse to cloud), a backend/cloud layer (traceability, predictive analytics, alerts, market recommendations), and two dashboards — one for infrastructure management and one for the farmer. Every entity is linked through a single traceability chain: Farmer → Product → Crate ID → Storage Location → Transport → Market.

2. System Architecture (Layered)

Firmware Layer — ESP32-C3 control loop on each crate

Communication Layer — wireless/GSM uplink + GPS

Backend / Cloud Layer — APIs, database, analytics, alert engine

Application Layer — Infrastructure Dashboard + Farmer Dashboard

3. Firmware Logic (On-Crate Control Loop)

Pseudocode outline of the ESP32-C3 control routine:

```
LOOP every N seconds:
  reading = SHT31.read()
  IF reading.temp within [setpoint - tolerance, setpoint + tolerance]:
    Peltier.setDutyCycle(MIN)
  ELIF reading.temp > setpoint + tolerance:
    Peltier.activate(FULL)
  ELIF approaching setpoint from above:
    Peltier.reduceDutyCycle()
  battery_soc = BMS.readSOC()
  runtime_estimate = estimateRuntime(battery_soc, duty_cycle_history)
  packet = {crate_id, reading, battery_soc, runtime_estimate, gps_location}
  send(packet) over comms module
  IF any threshold breached: triggerLocalAlert(LED/buzzer/OLED)
```

4. Data Pipeline

Sensor Reading (SHT31 / BMS / GPS) on crate

ESP32-C3 packages data packet

Wireless / GSM Transmission to cloud gateway

Backend Ingestion API validates & timestamps packet

Database Write (crate history, live status)

Analytics Engine (predictive runtime, alerts, price recommendation)

Dashboard APIs push updates to Infra & Farmer apps

5. Backend Modules

Module	Responsibility
Crate & Traceability Service	Maintains Farmer → Product → Crate ID → Storage → Transport → Market chain
Storage Management Service	Tracks grid/slot occupancy, entry/exit times, storage conditions
Transport Management Service	Assigns vehicles/crates, tracks route, ETA, delivery status
Predictive Analytics Engine	Estimates temperature-maintenance runtime vs remaining journey time
Alert Engine	Rule-based + threshold checks; routes alerts to farmer or infra team
Market Recommendation Service	Combines delivery data with price/demand feeds for selling suggestions
User & Access Management	Farmer accounts, infra-team roles, authentication

6. Predictive Battery & Runtime Estimation — Logic

Inputs consumed by the analytics engine:

- Battery SOC (from BMS)
- Current draw / power consumption history
- Ambient temperature (outside crate)
- Required setpoint temperature
- Peltier duty-cycle pattern
- Insulation characteristics (fixed per crate model)
- Historical consumption data for similar routes/conditions

Output: **Estimated Temperature-Maintenance Time Remaining**. The engine compares this against **Remaining Journey Time** (from route/ETA data). If maintenance time falls short of journey time, a critical alert is generated automatically.

7. Alert Engine — Rule Categories

Category	Trigger Conditions
Temperature	High / low deviation from setpoint; rapid rate of increase
Humidity	High / low humidity outside safe range
Battery	Low SOC; abnormal consumption; insufficient runtime for remaining journey
Device	Sensor failure; communication loss; controller/actuator fault

Routing: farmer-facing alerts are scoped to their own Crate ID; infrastructure-facing alerts are system-wide across the fleet.

8. Dashboard Data Flow

Infrastructure Management Dashboard consumes:

- Facility overview (slots, crates, dispatch status)
- Energy management (solar generation, charging status, consumption)
- Crate management (ID, position, status, SOC, sensor health)
- Transportation management (vehicles, routes, ETAs)
- System-wide alerts and analytics/reports

Farmer Dashboard consumes:

- My Crate (ID, product, quantity, status, location)
- Live condition (temperature, humidity, battery, runtime estimate)
- Tracking (live location, route, ETA, delivery progress)
- Notifications scoped to their own crate
- Market/delivery information and price recommendation
- Farmer history (past shipments, revenue, spoilage records)

9. Core Data Entities

Entity	Key Fields
Farmer	Farmer ID, name, contact, linked crates
Crate	Crate ID, product, quantity, current status, grid slot
Storage Slot	Grid/slot number, occupancy, required temp/humidity range
Sensor Reading	Crate ID, timestamp, temperature, humidity, SOC, location
Shipment	Crate ID, vehicle, route, dispatch time, ETA, delivery status
Alert	Crate ID, type, severity, timestamp, resolved status
Market Price	Market name, product, current price, demand indicator

10. Power-Mode Software Logic

The backend tags the facility's current operating mode and adjusts dispatch priority logic accordingly:

Mode	Software Behaviour
Grid Normal	Standard monitoring; prioritize charging crates ahead of scheduled dispatch
Solar Emergency	Dashboard flags 'SOLAR / EMERGENCY MODE'; non-essential loads disabled in control logic
Thermal Survival	System ranks most temperature-sensitive produce and auto-recommends priority dispatch; triggers alerts if outage exceeds safe duration